

How does a frost-driven price shock affect Ontario apple demand, and what is the carbon cost of import substitution?

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1 Abstract

This paper estimates the price elasticity of demand in Ontario's wholesale apple market and quantifies the carbon cost of frost-driven supply shocks. Using a county-year panel from OMAFRA covering 2010 to 2020, I estimate the demand elasticity by 2SLS, using annual frost days across Ontario counties as an instrument for farm-gate prices. The results suggest that wholesale demand is elastic: a 1% increase in price is associated with an approximately 1.96% decrease in marketed production.

Using this estimate, I conduct a counterfactual analysis based on a 10% frost-driven price increase. Assuming buyers fully substitute toward imports from Washington State, which accounts for roughly 80% of Canadian apple imports, the added cross-border transportation would generate an estimated 4,846 tonnes of CO₂ emissions annually. These findings identify an unmeasured climate externality arising from Canada's apple trade and support a policy of imposing carbon pricing on imported apples, and frost adaptation subsidies for Ontario growers.

2 Introduction

The paper focuses on Ontario apple market data. Ontario is Canada's largest apple-producing province, accounting for about 46% of national production ([1]). Ontario's apple-growing regions are mostly located along the shores of the Great Lakes where they have an advantage of having large water bodies that help moderate the extreme cold weather, thus protect the tender fruit from being exposed to damaging frost ([2]).

Though climate warming is projected to reduce total annual frost days in the Niagara Region from 125 to about 84 days by 2051–2080 ([2]), fewer frost days do not imply a lower frost risk during the growing season. First, warmer winters can cause apple trees to bloom about 10 days earlier under 2°C warming, while late frosts do not disappear as quickly, leaving blossoms be in the most vulnerable stage to freezing damage ([3], [4]). Second, warmer winters induce deacclimation, which reduces the cold tolerance of crops, making trees more susceptible to freezing temperatures ([5]). Frost damages apple yield through a biological mechanism: at full bloom, the flowers are at the most fragile state. For example, a temperature of -2.2°C can kill 10% of the flowers, and -3.9°C can kill 90% of them. When the ice forms inside the flower, the reproductive tissue dies, and the blossom is unable to set fruit, causing the overall production to reduce.

Across Ontario counties, the number of frost days varies year to year. I therefore use county-year variation in annual frost days as a proxy for the climate-driven supply risk of apple production. This motivates my research question: how does a frost-driven increase in Ontario apple prices affect marketed production quantity, and what additional CO₂ emissions result when Ontario buyers substitute lost local supply with imported apples? To quantify carbon dioxide emission of a frost shock in Ontario's apple market, I construct a counterfactual in which a frost

event raises the Ontario apple price by 10%. When local Ontario supply falls, buyers substitute toward imported apples. Since Washington State supplies approximately 80% of Canada's apple imports, I assume that lost Ontario supply is fully replaced by Washington imports trucked across the border, making it the basis for estimating the additional CO₂ emissions from cross-border transportation.

The paper is structured as follows. The data section constructs a county-level panel of Ontario apple farm-gate prices and marketed quantities, with annual frost days as the main instrument variable for supply side capturing county-level frost risk. The model section presents a structural log-linear supply and demand framework for the Ontario wholesale apple market and states the identifying assumptions. The estimation section reports four specifications: naive OLS, OLS with county fixed effects, 2SLS using frost days as the instrument, and 2SLS using Washington State farm-gate prices as a second instrument; it interprets the price elasticity of demand and discusses instrument validity. In the counterfactual analysis, I consider a hypothetical 10% frost-driven price shock to Ontario apple supply and quantify the additional CO₂ emissions released when buyers substitute toward Washington imports. Finally, the discussion addresses data limitations, the distinction between short-run and long-run equilibrium effects, and policy implications for managing climate-driven supply risk in Ontario apple production.

3 Data

I obtained annual apple production data from 1995 to 2024 across counties in Ontario from the Ontario Fruit and Vegetable Production dataset from OMAFRA ([6]). I then construct a county-level panel dataset with marketed production (in thousands of pounds) and average farm-gate prices (cents per pound). Marketed production is used as the quantity variable, and farm-gate price as the price variable. This captures the wholesale market, where buyers are apple distributors and wholesalers.

I restrict the sample to the 2010-2020 period, in which data all overlap, and to avoid long-run structural changes. The unit of observation is the county-year. The final panel contains 407 observations across 37 Ontario counties. Quantity, price, and annual frost days values are transformed by natural logarithms. The figure below is a plot of log marketed production against log farm-gate price across all observations. The red fitted line reflects a downward-sloping demand curve, and the widespread scatter at every price level reflects a heterogeneity in production capacity across counties, which motivates me to add county fixed effects in estimation specifications.

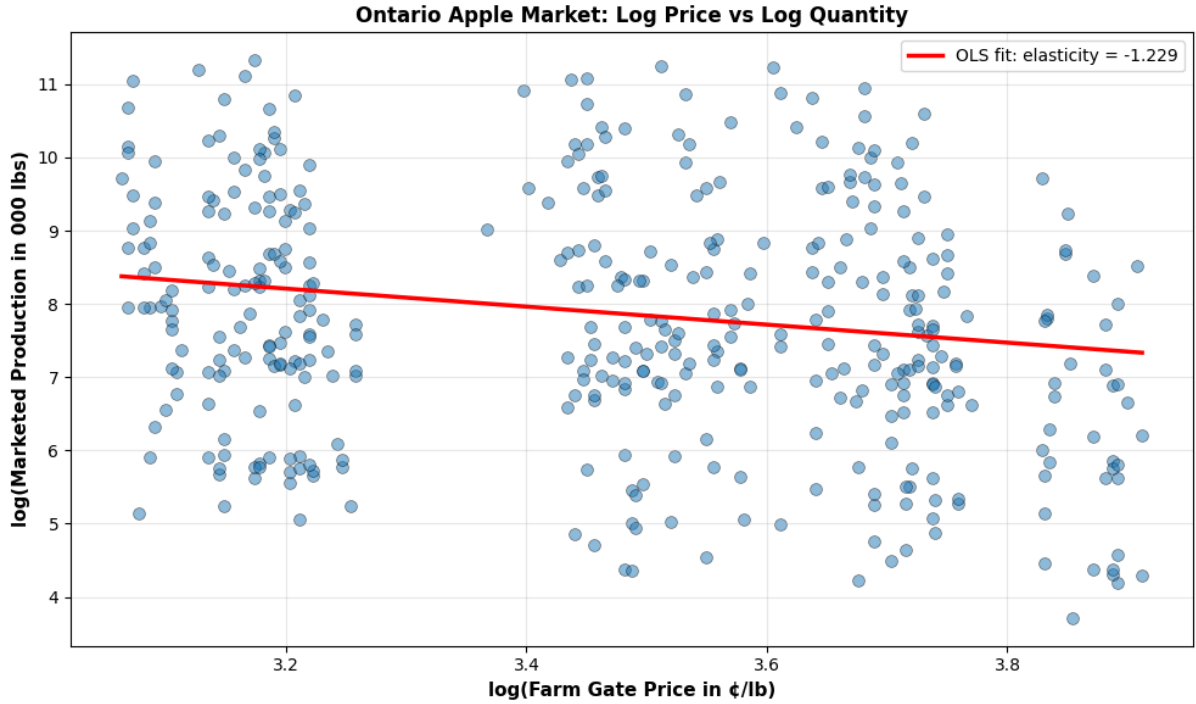


Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	N
Log_prod	7.739	1.668	3.714	11.325	407
Log_price	3.467	0.254	3.063	3.912	407
Log_wa_price (¢/lb)	3.658	0.167	3.355	3.920	407
Annual Frost Days (days/year)	132.288	19.745	70.400	188.000	407

4 Model

I model the wholesale market in Ontario, where buyers are processors and distributors who purchase apples directly from farmers. The key parameter of interest is the price elasticity of demand - how much demand of apple reduces when its price increases due to a late frost day that damages the total production. I assume the Ontario apple market is competitive: there are many growers and many buyers, all transact at a market-clearing price, so marketed production equals equilibrium quantity demanded. The model is estimated in the short run with orchard area fixed, so variations in prices and quantities reflect supply-side shocks within a given level of orchard capacity. Long-run adjustments such as orchard expansion and technological advances are not included in the model and considered as a limitation in my estimation.

Because price and quantity are jointly determined in the demand-supply structural model, OLS estimate would be biased as changes in price do not come purely from supply conditions. To estimate the demand elasticity of apple price, I therefore instrument Ontario farm-gate prices with the log of annual frost days in each county-year, z_{it} . Frost days are the number of days per year with daily minimum temperature below 0°C, obtained from the LAMPS Ontario Climate

Data Portal at York University [7]. In years with fewer frost days, which indicates a warmer winter and earlier bloom period, the probability of late spring frosts that damage apple blossoms is higher.

The instrument needs to satisfy two conditions. First, frost days must shift Ontario apple supply, and through supply, affecting the farm-gate price of apples. The channel is based on the biological mechanism, in which late spring frosts damage apple blossoms during the bloom window from late April to mid-May. Because annual frost days imperfectly capture spring-specific frosts, I treat it as a proxy for spring frost exposure. The IV thus might give measurement error, and underestimates the true responsiveness of demand which is likely more negative than the estimated one. Second, the timing of the last spring frost is unrelated to the demand of apples in the market. Late frost dates affect Ontario demand only through its effect on local supply and the corresponding market price. However, the main threat is that wholesale buyers can adjust purchases based on frost warnings before prices actually move, which would mean frost days might affect demand directly rather than only through supply. I assume that the buyers do not respond to the forecasting news of the spring frost and shift their demand before prices change due to reduced supply. The second assumption is a strong one and considered as limitations to the IV's validity.

As a robustness check, I add the log of Washington State apple price as a second instrument. The Washington State price instrument is relevant because if Washington prices rise up due to an unexpected harvest condition, then import competition, imported apples become more expensive, Ontario buyers shift toward local apples, and Ontario producers can raise farm-gate prices in response to stronger local demand. Ontario producers therefore gain more pricing power and can raise their own prices in response to stronger local demand. The exclusion restriction requires that Washington prices affect Ontario marketed production only through Ontario farm-gate prices, not through any direct effect on Ontario buyer demand. Ontario wholesalers observe local prices, not prices of Washington apples, so this assumption is plausible.

I specify a log-linear demand-supply system where σ is the price elasticity of demand and z_{it} is the number of annual frost days as a supply shock.

$$\begin{aligned}\log q_{it}^D &= \alpha^D - \sigma \log p_{it} + \epsilon_{it}^D \\ \log q_{it}^S &= \alpha^S + \beta^S \log p_{it} + \phi z_{it} + \epsilon_{it}^S\end{aligned}\tag{1}$$

- q_{it}^D : Marketed production (quantity demanded) in county i and year t
- q_{it}^S : Quantity supplied in county i and year t
- p_{it} : Ontario farm-gate price of apples in county i and year t
- z_{it} : Supply shifter-log of the annual frost days in county i and year t
- σ : Price elasticity of demand

- ϵ_{it}^D : Structural demand error term in county i and year t
- ϵ_{it}^S : Structural supply error term in county i and year t

$$\log q_{it}^D = \log q_{it}^S \quad (2)$$

$$\alpha^D - \sigma \log p_{it} + \epsilon_{it}^D = \alpha^S + \beta^S \log p_{it} + \phi z_{it} + \epsilon_{it}^S \quad (3)$$

$$\alpha^D - \alpha^S - \phi z_{it} + (\epsilon_{it}^D - \epsilon_{it}^S) = (\beta^S + \sigma) \log p_{it} \quad (4)$$

$$\log p_{it} = \frac{\alpha^D - \alpha^S}{\beta^S + \sigma} - \frac{\phi}{\beta^S + \sigma} z_{it} + \frac{\epsilon_{it}^D - \epsilon_{it}^S}{\beta^S + \sigma} \quad (5)$$

The first-stage reduced form is:

$$\log p_{it} = \pi_0 + \pi_1 z_{it} + v_{it} \quad (6)$$

•

$$\pi_0 = \frac{\alpha^D - \alpha^S}{\beta^S + \sigma} \quad (7)$$

•

$$\pi_1 = \frac{-\phi}{\beta^S + \sigma} \quad (8)$$

•

$$v_{it} = \frac{\epsilon_{it}^D - \epsilon_{it}^S}{\beta^S + \sigma} \quad (9)$$

5 Estimation results

Specification 1 estimates naive OLS with no controls, giving an elasticity of -1.496 (SE = 0.318, $p < 0.001$). The low R-squared value of 0.05 confirms that price alone explains little of the variation in production across counties, and the absence of controls means the estimate mixes both demand side and supply side effects so it should be treated as a biased baseline.

Specification 2 adds county fixed effects, reducing the estimated elasticity to -1.149 (SE = 0.086, $p < 0.001$). The sharp rise in R-squared value to 0.94 reflects that the large share of production differences can be explained by time-invariant county geography such as orchard area, soil quality, and proximity to the Great Lakes. The within-R-squared value of 0.33 confirms that price variation within a county over time explains 33% of year-to-year production changes, justifying county fixed effects as necessary controls. Even with fixed effects, the

Table 2: Estimatio results

	(1)	(2)	(3)	(4)
	OLS	OLS	IV	IVs
Constant	12.928*** (1.104)	9.603*** (0.356)	12.383*** (0.965)	7.580*** (0.471)
log(Price)	-1.496*** (0.318)	-1.149*** (0.086)	-1.956*** (0.273)	-0.561*** (0.123)
County FE	No	Yes	Yes	Yes
First-stage F	—	—	51.73	—
First-stage F (joint)	—	—	—	124.9
Observations	407	407	407	407
R^2	0.052	0.937	0.922	0.929
Adjusted R^2	0.050	0.931	0.914	0.922
F Statistic	22.19***	144.8***	114.4***	124.9***

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

For (2), within $R^2 = 0.326$.

For (4), Hansen J-statistic = 40.813, $p < 0.001$.

estimate, however, remains biased because joint determination of supply and demand causes simultaneity, which biases OLS estimate. I next use the IVs to address this simultaneity bias.

Specification 3 adds annual frost days, the number of days per year with minimum temperature below 0°C, as an instrument variable. The first-stage F-statistic of 51.7 confirms that frost days predict Ontario prices, and the estimated demand elasticity is -1.956 (SE: 0.273, $p < 0.001$), implying that a 10% increase in prices reduces marketed production by about 19.6%, all else equal. The magnitude of estimate from specification 3 is larger than the one obtained from specifications 1 and 2 because the instrument variable of frost days isolates the variations in prices due to supply shock from frost risks helps to identify the demand curve. In years with more frost days, the spring/winter seasons have longer periods of cold days, making the yield decrease, causing the apple prices to rise, and thus reducing the quantity demanded. The magnitude of the estimate -1.956 implies that the demand is quite elastic for Ontario apples. A plausible explanation is that wholesalers can easily switch to imported US apples which are direct substitutes in most processing and retail channels.

Specification 4 adds log Washington State apple price as a second instrument. The estimated demand elasticity falls in magnitude to -0.561 (SE = 0.123, $p < 0.001$), implying that a 10% rise in farm-gate prices reduces wholesale quantity demanded by about 5.6%. However, the Hansen J over-identification statistic is 40.8 ($p < 0.001$), rejecting the null that both instruments are jointly valid, meaning frost days and Washington prices capture different types of price shocks, and buyers respond to them differently. Annual frost days are specific to each county, thus identify the demand response along a narrow substitution margin: if Ontario county loses production

due to a late frost nigh, buyers can shift to neighboring counties or imports. When Washington prices rise, every Ontario county faces the same change at the same time, because Washington supply affects the whole province through the import channel, hence the substitution becomes expensive for all that buyers cannot easily switch away.

Since my counterfactual is a local frost shock to Ontario supply, I keep Specification 3, which uses only frost days, as the preferred estimate, and treat Specification 4 as a robustness check showing how the result changes with different instruments.

6 Counterfactual Analysis

I use the preferred elasticity from Specification 3 conduct the the counterfactual analysis. To translate the elasticity into a carbon externality, I assume that the reduction in Ontario apple consumption is fully replaced by imports from Washington State. This assumption is reasonable for the wholesale market because processors and distributors operate under fixed long-term contracts and cannot easily reduce the apple supplied; they substitute towards imports from foreign countries. Washington supplies approximately 80% of Canadian apple imports ([1]), making it the most likely substitution destination.

Consider a 10% increase in Ontario farm-gate prices triggered by a severe spring frost that damages apple blossoms during bloom. There are four assumptions along with the below calculations. First, the price shock is entirely supply-driven — frost damages blossoms and contracts local harvest, with no shift in buyer demand. Second, lost Ontario supply is fully replaced by Washington imports, with no substitution toward other fruits or other import origins. Third, imported apples are transported from Yakima, Washington to Toronto, a one-way road distance of approximately 3,500 km. Fourth, I apply the ECCC freight emission factor of 62 g CO₂ per tonne-kilometre. These assumptions likely underestimate the true carbon cost because packaging, cold storage, and cross-border logistics are excluded. In practice, buyers also substitute toward other fruits, which would reduce the volume of Washington imports below the full-replacement estimate. The calculation therefore might serve as an upper bound on transport emissions from the substitution channel.

I apply the estimated elasticity of -1.956 to compute the quantity reduced in Apple production:

$$\Delta Q = \exp(-1.956 \times 0.0953) - 1 \approx -0.170 \quad (10)$$

At the county level, the average marketed production is approximately 7,816,000 lbs per year. A 17% reduction gives $7,816,000 \times 0.170 = 1,329,490$ lbs of apples lost due to the frost shock. In tonnes, this is $1,329,490 \times 0.000454 = 604$ tonnes per county. Across all 37 apple-growing counties in Ontario, the total production loss is $604 \times 37 = 22,333$ tonnes per year.

Locally distributed Ontario apples typically travel less than 200 km to retail markets, while apples imported from Washington need to travel over 3500 km. Applying the ECCC freight emission factor of 62 g CO₂ per tonne-kilometre, I computed the additional transport emissions to be: $22,333 \times 3,500 \times 0.000062 = 4,846$ tonnes of CO₂ per year.

The 4,846 tonnes of CO₂ is the result of the large difference in transport distance between local and imported apples. If the amount of imported apples is even smaller, the carbon emissions per tonne of substituted apples would still remain high because they must be transported over such long distances. The counterfactual suggests that unexpected frost shocks cause hidden carbon cost to Ontario's apple market, which depends on the severity of the supply disruption and the reliance on imported apples when local supply is affected by extreme weather.

7 Conclusion and Policy Recommendation

This paper estimates the price elasticity of demand in Ontario's wholesale apple market using a county-year panel from 2010 to 2020, with the assumption that annual frost days isolate supply side price variation independent of buyer demand. The preferred IV specification estimates a demand elasticity of -1.956, indicating that a 1% increase in wholesale price reduces marketed production by approximately 1.96%. Applying this estimate to a frost-driven 10% price shock, Ontario loses approximately 22,333 tonnes of apple supply per year across 37 counties. At the higher price due to the shock, Ontario buyers reduce their quantity demanded of Ontario apples by 19.6%. When buyers substitute local apples with Washington imports, the additional transport emissions are 4846 tonnes of CO₂ per year. These results represents a measurable climate cost linked to frost risk in Canadian fruit production.

Several limitations are about data measurement and assumption violation. First, marketed production includes apples drawn from cold storage, so the timing of observed quantities may not align precisely with the year of production. Second, the frost-days instrument may also be correlated with long-run technological change that improved frost protection and more resilient apple varieties - all of those factors would violate the exclusion restriction. I treat the IV estimates as plausibly causal given the first-stage strength and the biological mechanism of how frost damage apple blossoms, but there should be a need of extra estimates for comprehensiveness.

An additional caveat is about the time horizon of the estimated elasticity. Because the model holds orchard area fixed, the elasticity of -1.956 reflects only the short-run demand response to a price shock within a given level of orchard capacity. In the long run, farmers can adjust planted area in response to sustained price or climate forecasts: with more unexpected frost-free days due to climate change could make farmers to reduce plantings, exit the market, or invest in more frost-resilient cultivars. The long-run elasticity would therefore reflect both demand-side substitution and supply-side adjustment, and would likely differ in magnitude from the short-run estimate in this paper. A more comprehensive model would include controls for orchard area to adjust over time, input costs, weather conditions, and other macroeconomic factors. This would

reduce omitted-variable bias and isolate the frost-driven supply channel from other factors that simultaneously shift Ontario's apple supply and demand.

The policy implication of these results is based on two channels. First, frost-driven supply shocks reduce Ontario apple production, suggesting that the Government of Ontario should subsidize adaptation technology such as wind machines, frost fans, and production insurance([8]), to lower the the frequency and severity of yield losses due to frost days. Second, when residual frost shocks still occur, under assumption that buyers have to substitute toward Washington imports and generate additional 4,846 tonnes of CO₂ as a negative externality, a carbon tax or border carbon adjustment on imported fresh produce would compensate for it. The tax might also be able to redirect the substitution toward cold-storage inventory, or other fruits.

Each policy addresses a different cost: a subsidy on adaptation technique reduces the frost damage, while carbon pricing internalizes the emissions cost of import substitution. However, this recommendation is based on strong assumptions about the validity of the frost-day instrument and full substitution toward Washington imports, which may overstate the carbon cost. Therefore, I would extend this analysis in future work by adding supply controls, and by modeling the substitution incentives of buyers and fruit categories to better capture the hidden carbon cost.

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